Lec 2. · Exact eqn.

· separation variable.

· Integrating factor [optional]

Section 2.6 [BDM]

- 1^{st} order ODE

$$\frac{dy}{dt} = f(t, y).$$

- General form

$$M(x, y) + N(x, y)\frac{dy}{dx} = 0 \quad ①$$

- Separable:

$$M = M(x), \quad N = N(y).$$

- Q: More general case ??

[I]. ① is solvable if $\exists\ \phi(x, y)$ s.t.

chain rule.

$$\frac{d}{dx}\phi(x, y(x)) = \partial_1\phi(x, y) + \partial_2\phi\frac{dy}{dx}$$

- If $\exists \begin{cases} \partial_1\phi(x, y) = M(x, y) \\ \partial_2\phi(x, y) = N(x, y) \end{cases}$ ③, then ① Exact differential eqn.

In this case:

$$\frac{d}{dx}\left[\phi(x, y(x))\right] = 0$$

$$\Rightarrow \phi(x, y(x)) = C. \quad ②$$

Further solve $y(x)$ from ②.

Q: How can we check if ① is exact?

Thm: Let M, N, $\partial_2 M(x,y), \partial_1 N(x,y)$ be continuous in $(x,y) \in (a,b) \times (c,d) \doteq R$

① is exact if and only if

$$\partial_2 M(x,y) = \partial_1 N(x,y), \quad \forall (x,y) \in R. \qquad ④$$

Pf: $\boxed{\text{"}\Rightarrow\text{"}}$ Exact $\Rightarrow \exists\, \phi$ s.t.

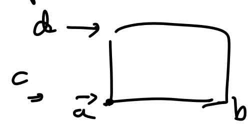
$$\partial_1 \phi(x,y) = M, \quad \partial_2 \phi(x,y) = N. \qquad \text{(see ③)}$$

Mixed derivative match

$$\Rightarrow \partial_2 M = \partial_2 \partial_1 \phi = \partial_1 \partial_2 \phi = \partial_1 N$$

$\Rightarrow$ ④

$\boxed{\text{"}\Leftarrow\text{"}}$ If ④ holds, want to find ϕ with ③



First achieve $\partial_1 \phi(x,y) = M(x,y)$ by , $x_0 \in (a,b)$.

$$\phi(x,y) = \int_{x_0}^{x} M(s,y)\, ds + h(y).$$

Next: want $\partial_2 \phi(x,y) = N(x,y)$

$$\Leftrightarrow \int_{x_0}^{x} \partial_2 M(s,y)\, ds + h'(y) = N(x,y)$$

Using ④ $\partial_2 M(x,y) = \partial_1 N(x,y)$ in R.

$$\text{LHS} = \int_{x_0}^{x} \partial_1 N(s,y)\, ds + h'(y).$$

$$= N(x,y) - N(x_0,y) + h'(y).$$

[Fundamental theorem of Calculus].

LHS = RHS if $h'(y) = N(x_0, y)$.

so take $h(y) = \int_{y_0}^{y} N(x_0, t)\,dt$.

$$\phi(x,y) = \int_{x_0}^{x} M(s,y)\,ds + \int_{y_0}^{y} N(x_0,t)\,dt.$$

E.g.

$$M(x) + N(y)\frac{dy}{dx} = 0$$

Separable: $\partial_y M(x) = 0 = \partial_x N(y)$.

$$[y + M(x)] + [x + N(y)]\frac{dy}{dx} = 0$$

Exact.

$$y^2 x + (1 + x^2 y)\frac{dy}{dx} = 0$$

$$\partial_y [y^2 x] = 2xy = \partial_x [1 + x^2 y]$$

Define

$$\phi(x,y) = \int_{x_0}^{x} s y^2\,ds + h(y)$$

$$= \frac{x^2}{2} y^2 + \tilde{h}(y)$$

$$\partial_y \phi = x^2 y + \tilde{h}'(y) = x^2 y + 1 \Rightarrow \tilde{h} = y$$

$$\Rightarrow \phi(x,y) = \frac{x^2 y^2}{2} + y$$

Thus

$$\frac{d}{dx}\phi[x, y(x)] = 0$$

$$\frac{x^2 y^2}{2} + y = C. \quad \text{Solve } y.$$

II: Integrating factor.

$$M(x,y) + N(x,y)\frac{dy}{dx} = 0 \qquad (1)$$

$\forall\ \mu(x,y) \neq 0$, (1) $\iff$

$$(\mu \cdot M)(x,y) + (\mu N)(x,y)\frac{dy}{dx} = 0$$

If

$$\partial_2[\mu M] = \partial_1[\mu N]. \quad (1) \text{ can be made exact.}$$

(2)

Derive conditions for μ

Warning: In general, (2) is a PDE!

Try $\mu = \mu(x)$ or $\mu = \mu(y)$!!